Microsoft Elevate

CAPSTONE PROJECT

Road Accident Analysis Using Power BI

PRESENTED BY

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OUTLINE:

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- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
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PROBLEM STATEMENT:

- Road accidents are a major public safety issue causing injuries, fatalities, and economic loss every year. It is difficult for authorities to understand accident patterns without proper data analysis.
- Large volumes of accident data contain information about **road type, weather conditions, speed limits, light conditions, and accident severity**, but this data is not easily interpretable in raw form.
- Therefore, there is a need for a **data visualization and analysis system** that can help identify accident patterns, risk factors, and trends to support better road safety decisions.

PROPOSED SOLUTION:

The proposed system is a **Road Accident Analysis Dashboard using Power BI** that transforms raw accident data into interactive visual insights.

The system performs the following:

Data Collection

- Accident dataset containing accident severity, road type, weather conditions, speed limits, and location.

Data Visualization

- Interactive dashboards
- Charts for accident severity
- Weather and light condition analysis
- Road surface and road type analysis
- Location-based accident mapping

User Interaction

- Filters for weather conditions, road type, speed limit, light condition
- This helps users easily explore accident patterns and risk factors.

SYSTEM APPROACH:

System Requirements

- Computer/Laptop
- Power BI Desktop
- Accident dataset (CSV/Excel format)

Technologies Used

- **Power BI** – for dashboard development
- **Microsoft Excel / CSV Dataset** – for accident data
- **Power Query** – for data transformation

Visualizations Used

- KPI Cards
- Pie Charts
- Bar Charts
- Column Charts
- Map Visualization
- Filters (Slicers)

ALGORITHM & DEPLOYMENT:

Data Input

- The system takes accident data including:
- Accident severity, Road type, Speed limit, Weather condition, Light condition, Road surface condition, Accident location

Data Processing Steps

- Import dataset into Power BI
- Clean and transform data using Power Query
- Build visualizations for analysis

Deployment

- The dashboard is developed in **Power BI Desktop**
- It can be published to **Power BI Service** for sharing and online access.

RESULT:

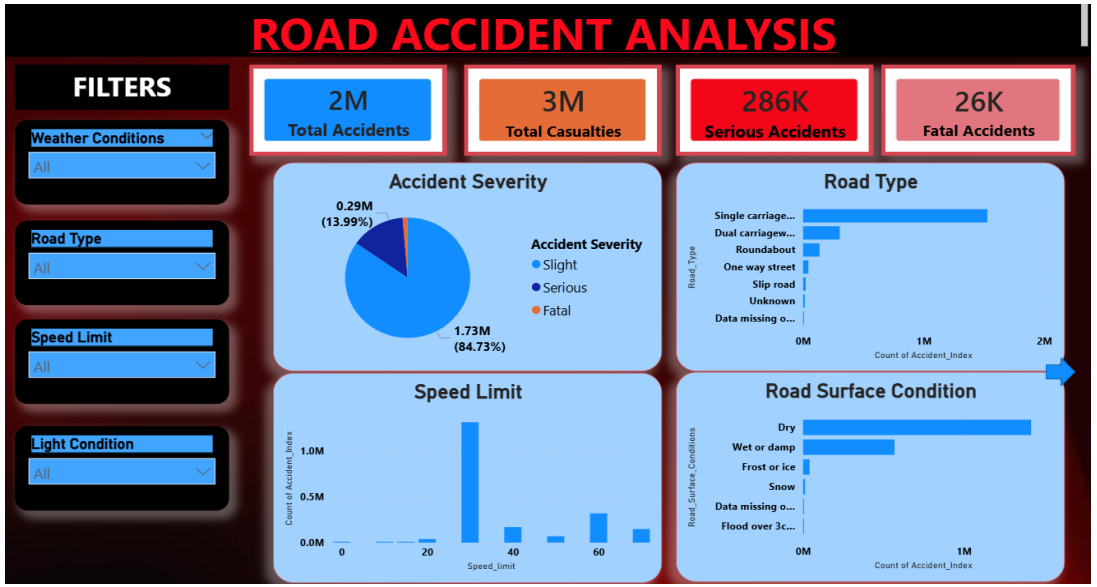
- The developed dashboard provides clear insights into road accident patterns.

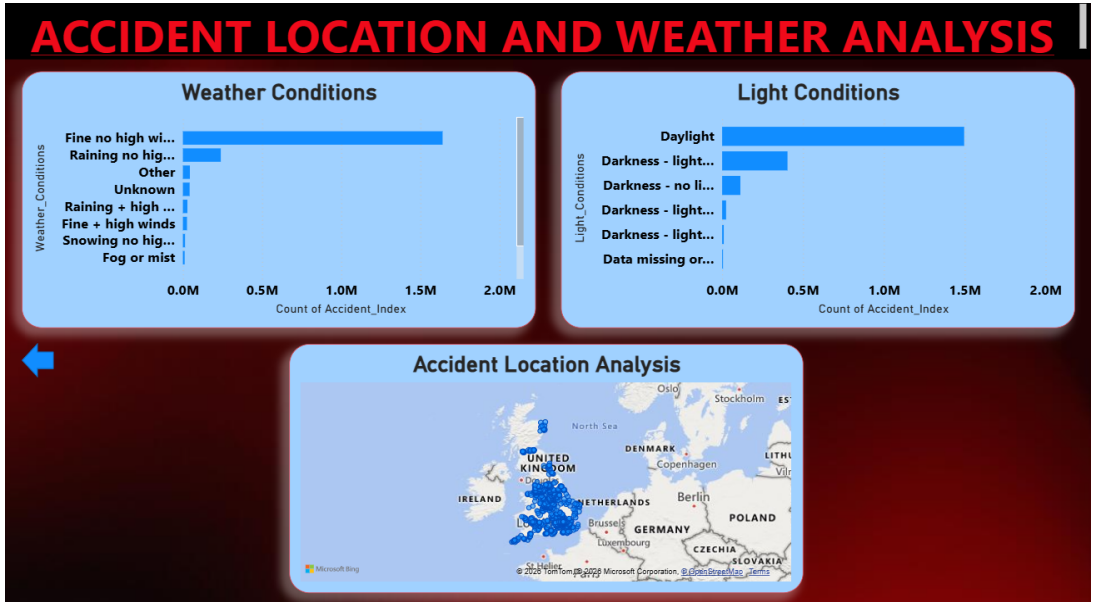
Key Insights from Dashboard

- **Total Accidents:** 2 Million
- **Total Casualties:** 3 Million
- **Serious Accidents:** 286K
- **Fatal Accidents:** 26K

Analysis Performed

- Accident severity distribution
- Road type accident comparison
- Speed limit accident analysis
- Road surface condition impact
- Weather condition analysis
- Light condition analysis
- Accident location visualization on map
- The dashboard allows users to explore data interactively using filters





CONCLUSION:

- The Road Accident Analysis dashboard helps understand accident patterns through interactive visualization.
- It highlights important factors affecting accidents such as **weather conditions, road type, speed limits, and light conditions**.
- This analysis can help government authorities and traffic departments make **data-driven decisions to improve road safety and reduce accidents**.

FUTURE SCOPE:

Future improvements can include:

- Real-time accident data integration
- Predictive analysis using machine learning
- Integration with traffic monitoring systems
- Accident hotspot prediction
- Mobile dashboard access
- AI-based accident risk prediction

REFERENCES:

- Power BI Documentation
- Microsoft Learn – Data Visualization
- UK Road Safety Accident Dataset -Kaggle
- Data Analytics and Visualization Tutorials

GitHub Link: https://github.com/Shamitha777/Road-Accident-Analysis-Power_BI.git

Thank You